



Swiss Institute of
Bioinformatics

INTRODUCTION TO SPATIAL TRANSCRIPTOMICS DATA ANALYSIS

QC (sequencing-based)

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June 03-05, 2026

Which QC do you use for single-cell data?

QC for Visium HD is conceptually the same as single-cell QC – but computed *per High-Definition spot* rather than per cell.

Cell → Spot

Why QC of spatial data?

High-density spots can capture technical artifacts (folds, bubbles, staining issues).

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Poor-quality regions can bias downstream analyses (clustering, spatial DE, trajectory analysis).

QC in Visium HD

Same metrics as single-cell RNA-seq:

- Total UMIs per spot

- Number of genes per spot

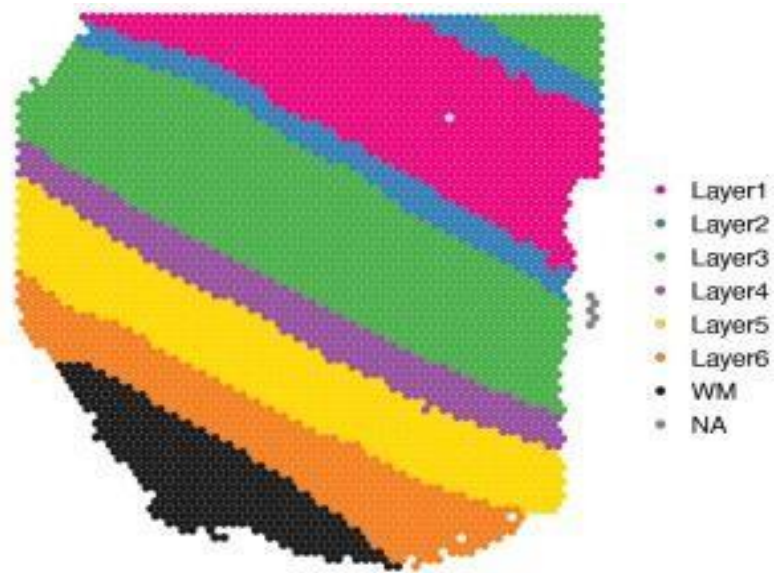
- Mitochondrial fraction

Key difference: unit is HD spot, not single cell.

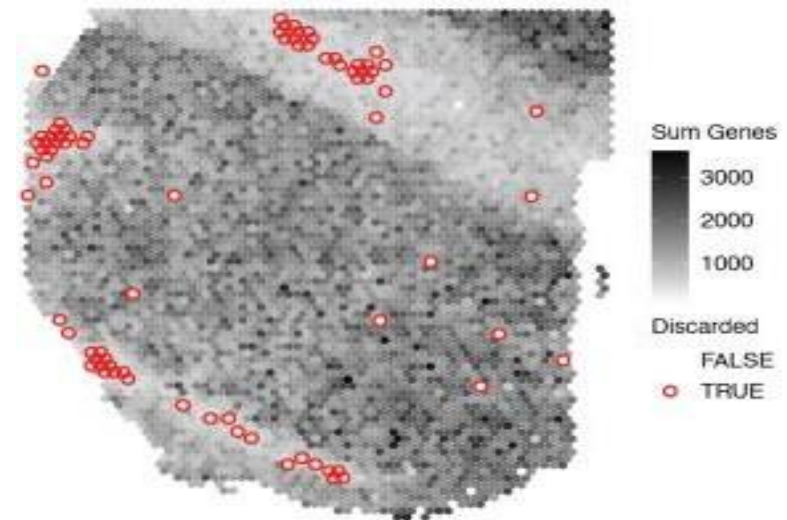
Neighboring spots share microenvironment → spatial artifacts can emerge

Example of spots removal

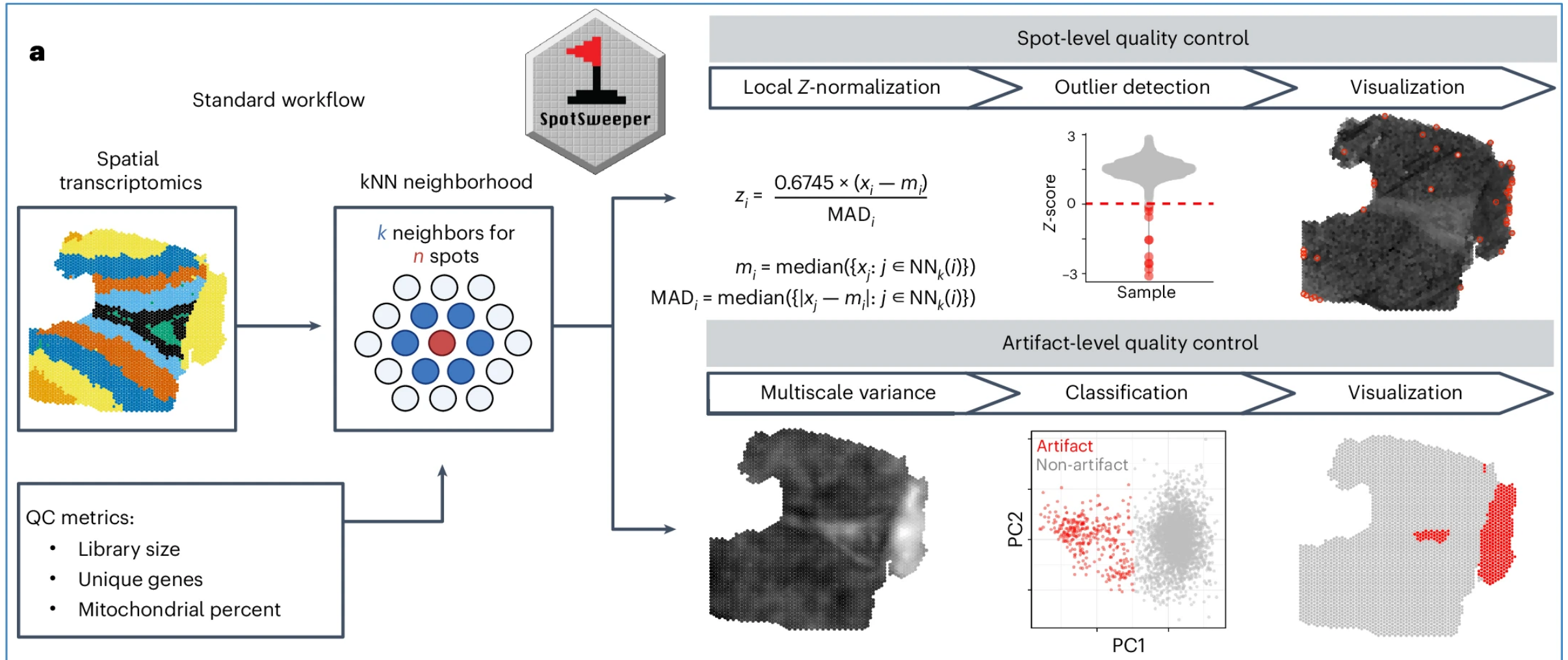
Annotation



Global

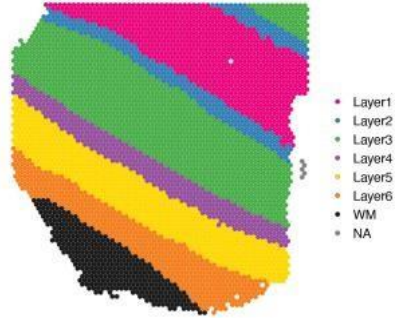


SpotSweeper: spatially aware QC

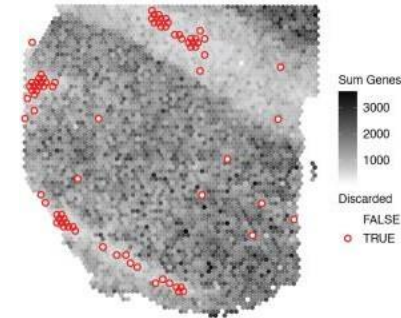


SpotSweeper (spot-level artifacts)

Annotation

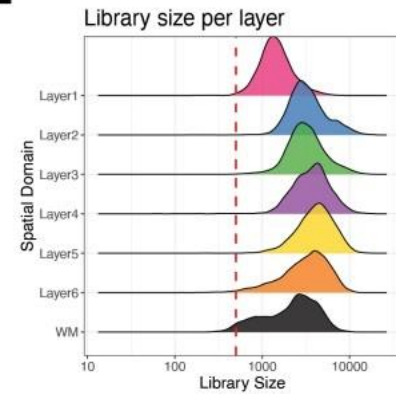


Global

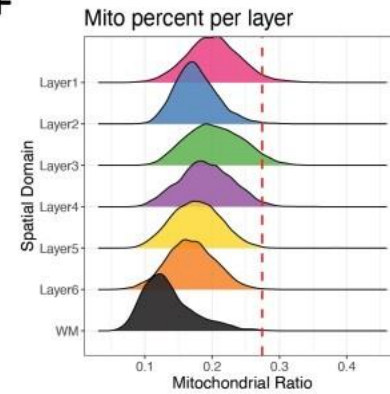


Global outliers

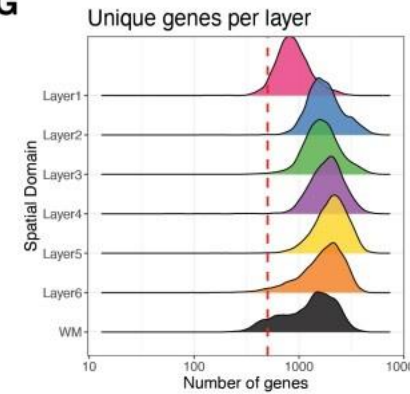
E



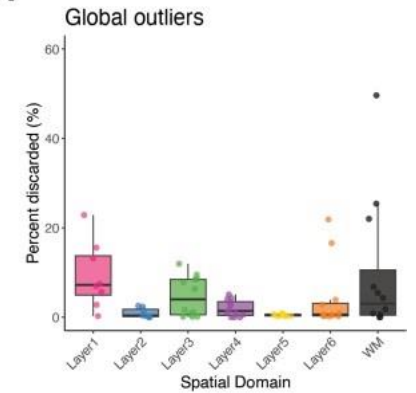
F



G

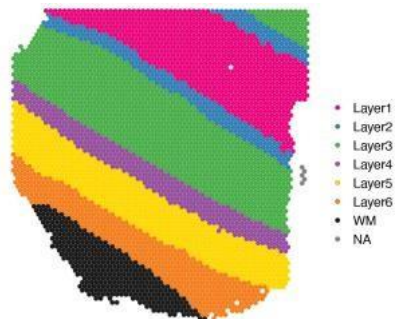


H

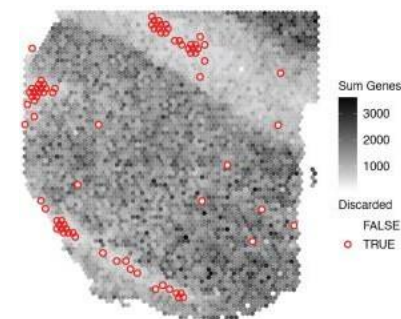


SpotSweeper (spot-level artifacts)

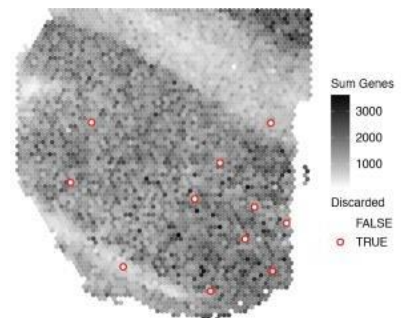
Annotation



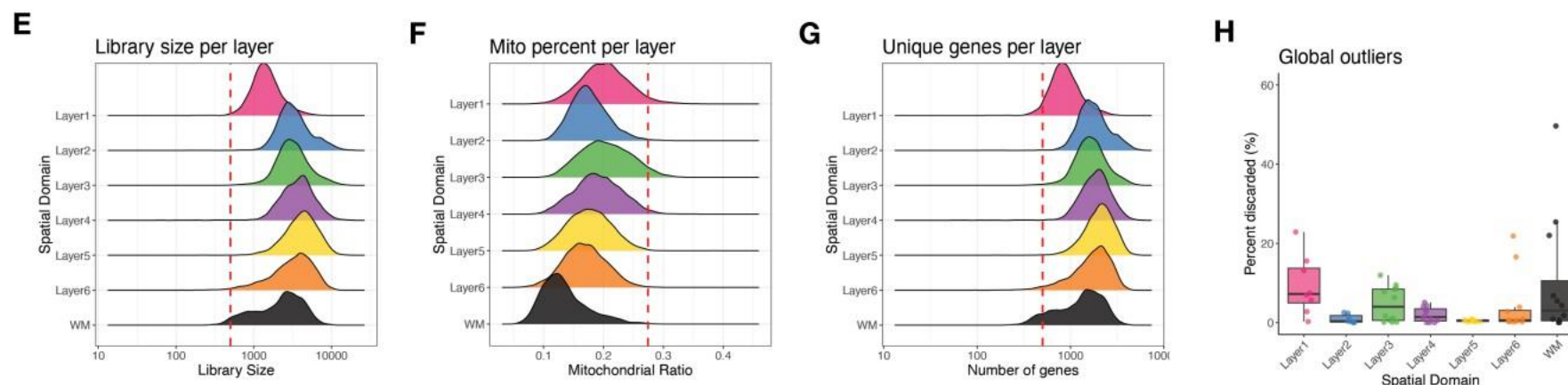
Global



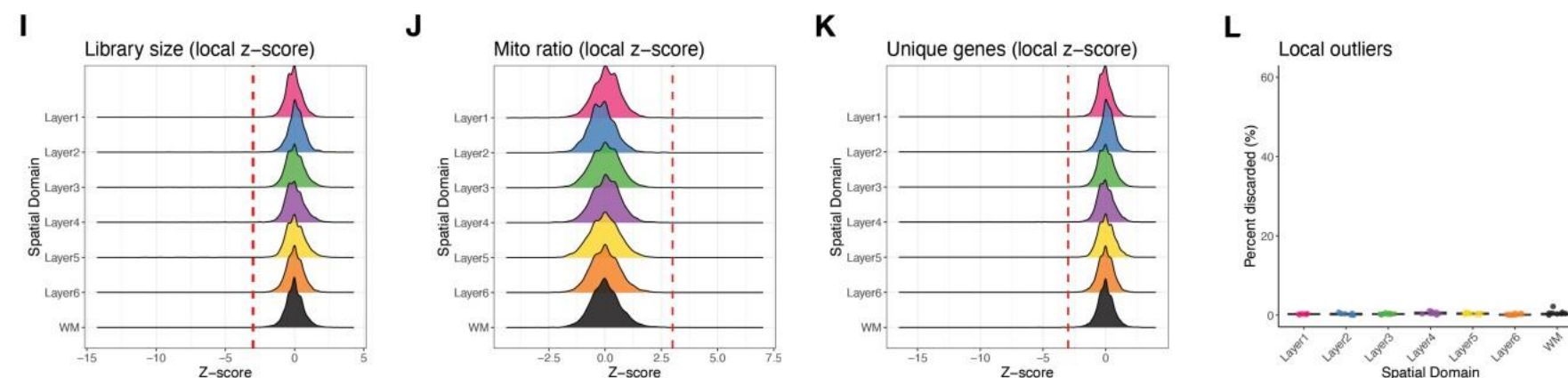
Local



Global outliers



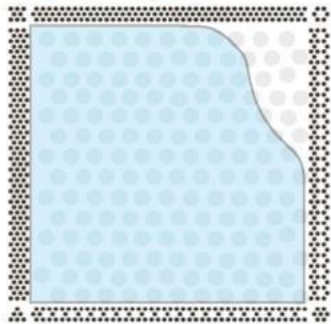
Local outliers



SpotSweeper (region-level artifacts)

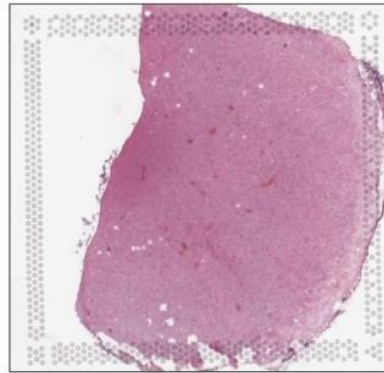
Incomplete coverage of Visium array

Liquid reagent

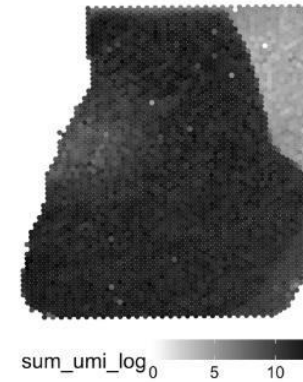


Dry spots result in
smaller library size
and fewer genes
detected

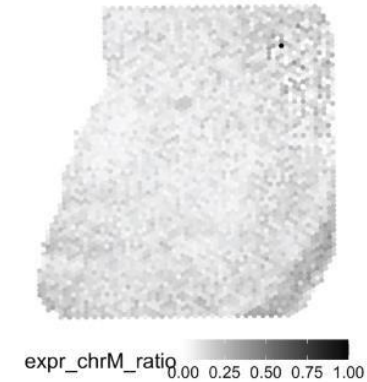
Br3942_mid



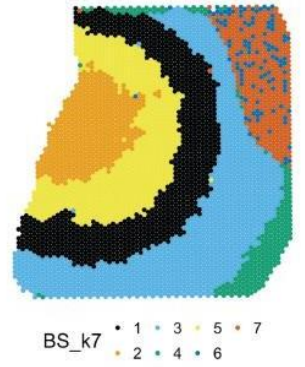
Library size



Mito Ratio



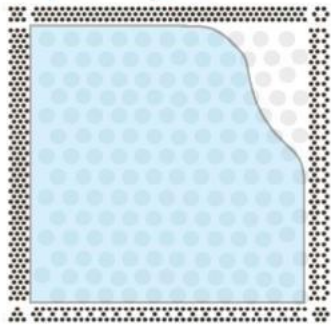
BayesSpace k=7



SpotSweeper (region-level artifacts)

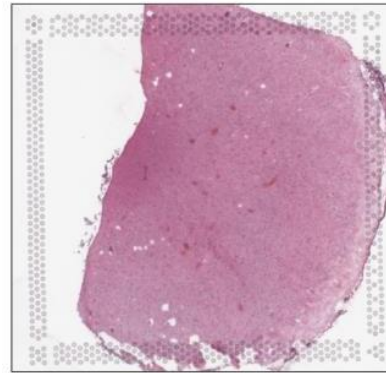
Incomplete coverage of Visium array

Liquid reagent

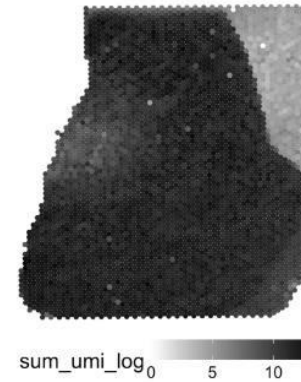


Dry spots result in smaller library size and fewer genes detected

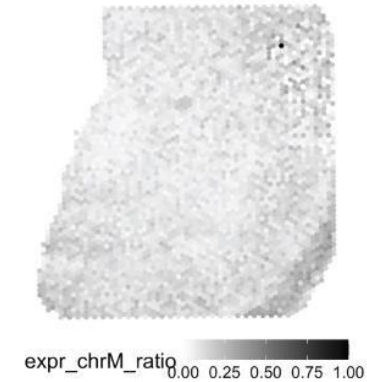
Br3942_mid



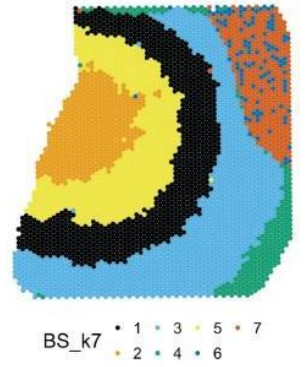
Library size



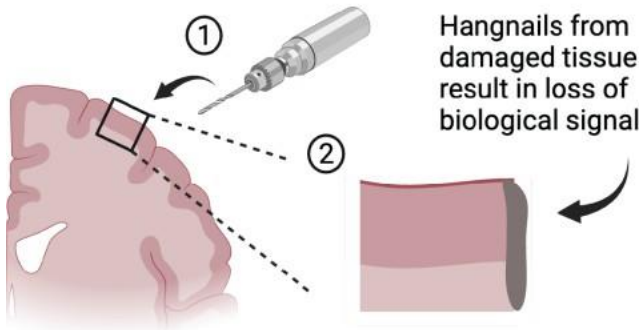
Mito Ratio



BayesSpace k=7



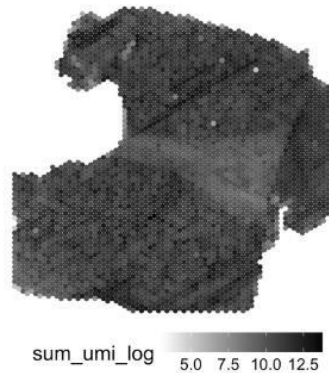
Tissue damage during dissection



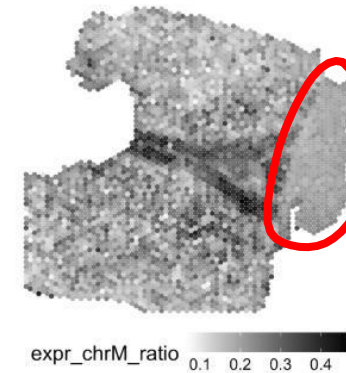
Br8325_ant



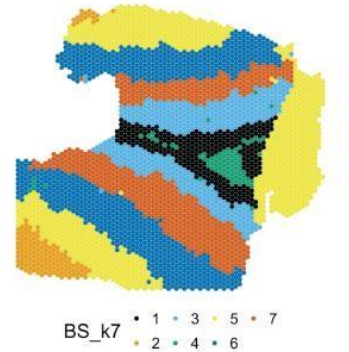
Library size



Mito Ratio

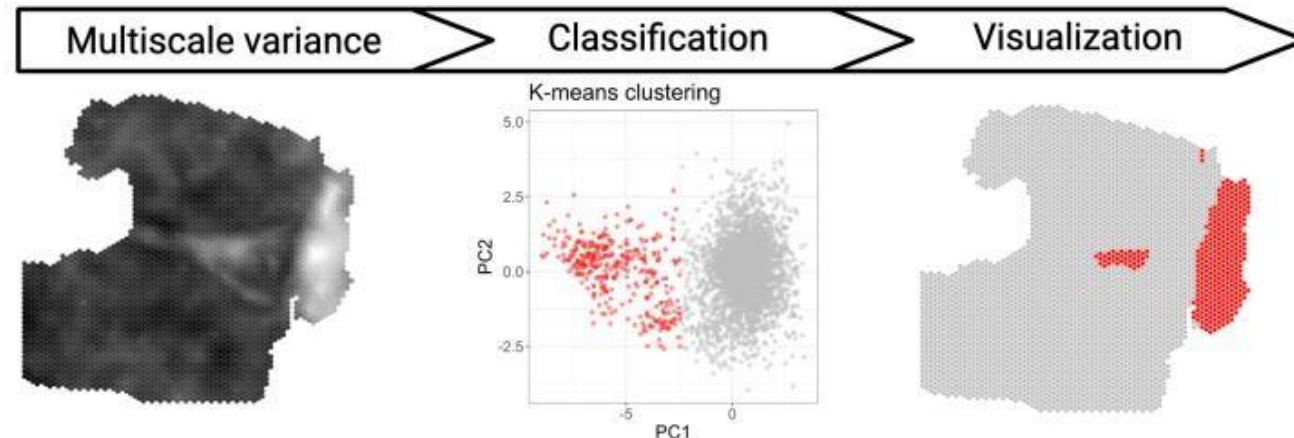


BayesSpace k=7



SpotSweeper (region-level artifacts)

1. The k-NN for each spot are identified based on the spatial coordinates
2. For each neighborhood size (i.e., scale), local variance of the mitochondrial ratio is calculated and adjusted for a mean-variance relationship using linear regression
3. Mean-corrected local variance = the residuals of the linear regression
4. Perform PCA on the mean-corrected local variances of all neighborhood sizes
5. Apply k-means clustering (k=2) in the first two PCs to identify regional artifacts compared to high-quality tissue



Preserve biological variability while removing technical artifacts: **SpotSweeper**

Detects **spatially coherent low-quality regions** in Visium HD slides.

Uses **per-spot QC metrics** (mainly mitochondrial fraction) and **spatial neighborhood info**.

Flags artifacts for downstream masking/removal.

Works across **multiple neighborhood scales** to detect both small and large defects.

How do you normalize your single-cell data?

Log-normalization

$$Y_{ij} = \log_e \left(\left(\frac{X_{ij}}{\sigma_i X_{ij}} \right) + 1 \right)$$

- Simplest and most commonly-used normalization strategy
- Divide all counts for each spot by a spot-specific scaling factor (i.e. size factor)
- Assumes that any cell-specific bias (e.g., in capture or amplification efficiency) affects all genes equally via scaling of the expected mean count for that cell

How about any other method?

How about any other method?

Any method from single-cell field?

Can we use scTransform?

Bhuva et al. *Genome Biology* (2024) 25:99
<https://doi.org/10.1186/s13059-024-03241-7>

Genome Biology

SHORT REPORT

Open Access

Library size confounds biology in spatial transcriptomics data



Dharmesh D. Bhuva^{1,2,3*}, Chin Wee Tan^{2,3,4}, Agus Salim^{2,5}, Claire Marceaux^{3,6}, Marie A. Pickering⁷, Jinjin Chen^{2,3}, Malvika Kharbanda^{1,2,3}, Xinyi Jin^{2,3}, Ning Liu^{1,2,3}, Kristen Feher^{1,2,3}, Givanna Putri^{2,3}, Wayne D. Tilley⁷, Theresa E. Hickey⁷, Marie-Liesse Asselin-Labat^{3,6}, Belinda Phipson^{2,3†} and Melissa J. Davis^{1,2,3,4,8†}

- Tested the effects of normalization on spatial domain identification

Though sctransform removes library size effects effectively, their confounding with biology results in removal of biological effects as well.

See also for imaging-based SRT...

Atta et al. *Genome Biology* (2024) 25:153
<https://doi.org/10.1186/s13059-024-03303-w>

Genome Biology

RESEARCH

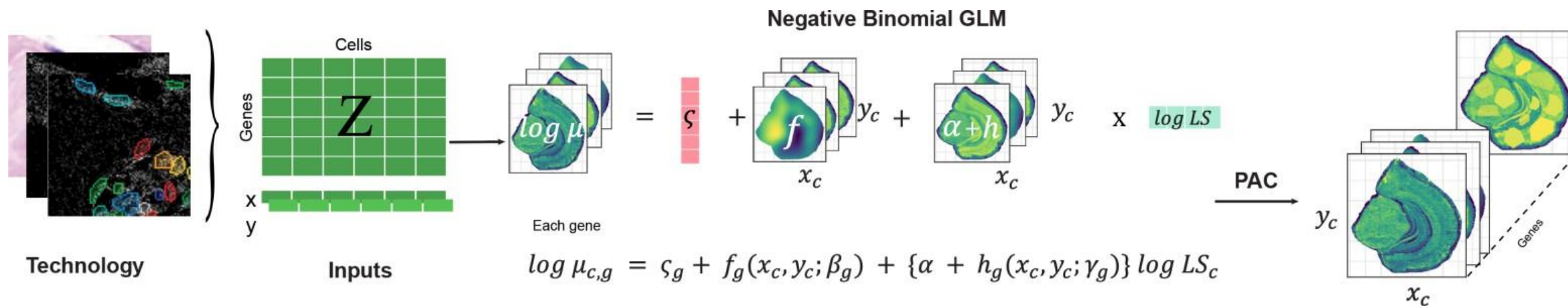
Open Access

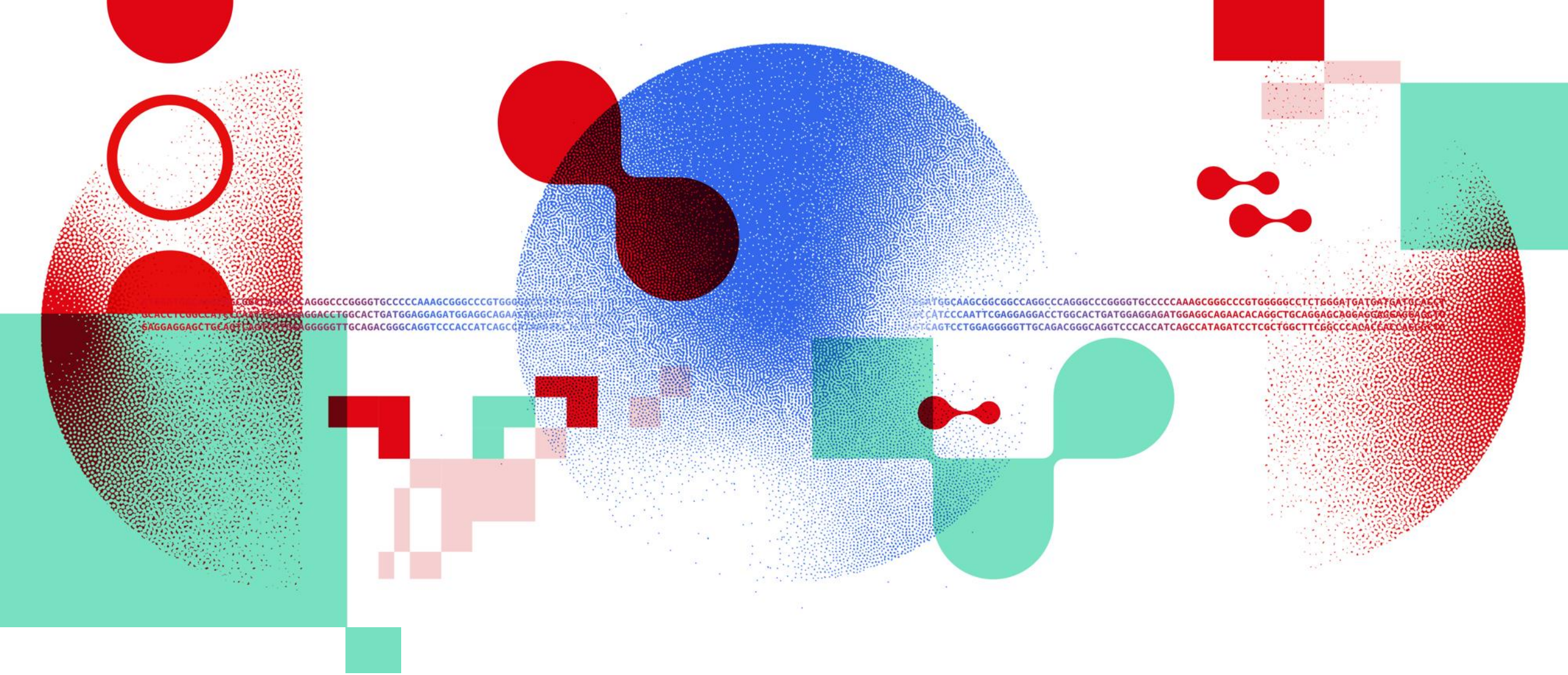
Gene count normalization in single-cell imaging-based spatially resolved transcriptomics



Lyla Atta^{1,2}, Kalen Clifton^{1,2}, Manjari Anant^{2,3}, Gohta Aihara^{1,2} and Jean Fan^{1,2*}

SpaNorm





Thank you

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